

Jacob Silva

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EXPERIENCE

Production Software Engineer Internship

June 2026 - Nov. 2026

Apple

San Diego, CA

- Build deployment and stage management tooling in **Swift** that teams across Apple use to ship **Kubernetes** services, covering **30+ repositories** and hundreds of commits a day that previously needed image tags updated by hand.
- Own the internal deployment dashboard, including a rebuilt interface and an analytics view that turns raw pipeline history into failure trends, so customer teams catch failed rollouts sooner.
- Ship stage management primitives in **server-side Swift** that let teams promote builds through deployment stages without hand-editing cluster configs.

QA Automation Software Engineer Internship

Jan. 2025 - July 2025

Apple

San Diego, CA

- Cut team debugging time **30-40%** by building a Chaos Client on **ToxiProxy** that simulates network failures, latency spikes, and service degradation against **distributed systems** in **continuous integration**.
- Designed **fault injection** scenarios that exposed dependency failures and latency bottlenecks ordinary testing had never reached.

Technical Specialist

Oct. 2025 - Present

Apple Retail

La Jolla, CA

- Diagnose and repair hardware, software, and account issues across the Apple lineup, with consistent **first-contact resolution** and satisfaction ratings in the **90th percentile** of stores.
- Reduce follow-up cases and customer wait times through troubleshooting education and streamlined check-in and repair coordination during high-traffic hours.

iOS/macOS Support Advisor

Jan. 2024 - Oct. 2025

Apple Support College Program

San Diego, CA

- Resolved **iOS** and **macOS** hardware, software, and account-level issues via phone support, escalating complex infrastructure issues for root-cause investigation while maintaining **above-average satisfaction ratings** through proactive customer education.

PROJECTS

Cartridge | *Swift, SwiftUI, Core Animation, AVAudioEngine*

- Wrote a Game Boy and Game Boy Color emulator in **Swift**, covering the CPU, graphics, sound, and memory bank controllers, running on **iOS** and **macOS** with save states and **iCloud** sync.
- Drove both platforms from one shared **Swift package**, holding full hardware speed with a lock-free audio ring buffer feeding **AVAudioEngine** and a **Core Animation** layer for frame output.

Sift | *Swift, SwiftUI, PhotoKit*

- Built an **iOS** app that finds near-duplicate photos using **perceptual hashing** behind a band index, avoiding the quadratic comparison a naive scan requires.
- Scans roughly **1,000 photos per second**; found 1,087 redundant photos across 652 groups in a 10,500-photo library.

Barbell | *SwiftUI, SwiftData, watchOS, HealthKit, ActivityKit*

- Built an **iOS** and **watchOS** lifting tracker driven by a rotation rather than a calendar, with **CloudKit** sync and a Live Activity rest timer on the Lock Screen.
- Added an Apple Watch companion with live heart rate through **HealthKit** workout sessions.

EDUCATION

San Diego State University

San Diego, CA

Bachelor of Science in Computer Science, Minor in Statistics, Minor in Mathematics

May 2026

TECHNICAL SKILLS

Languages: Swift, Python, Java, JavaScript, C#

iOS & Apple Frameworks: SwiftUI, SwiftData, WidgetKit, Swift Charts, CloudKit, HealthKit, ActivityKit, PhotoKit, Core Animation, AVFoundation

Testing: Swift Testing, XCTest, XCUITest

Backend & Infrastructure: Server-side Swift, Kubernetes, Docker, REST APIs, CI/CD, GitHub Actions, distributed systems

Tools & Data: Xcode, Instruments, Git, PyTorch, NumPy, Pandas, FastAPI